

Question:

The 9 members of a baseball team went to an ice-cream parlor after their game. Each player had a singlescoop cone of chocolate, vanilla, or strawberry ice cream. At least one player chose each flavor, and the number of players who chose chocolate was greater than the number of players who chose vanilla, which was greater than the number of players who chose strawberry. Let \$N\$ be the number of different assignments of flavors to players that meet these conditions. Find the remainder when \$N\$ is divided by \$1000\$.

Reasoning + Answer:

<think> To solve the problem , we need to find the number of ways to distribute 9 ice cream flavors among 9 players with the constraints that the number of players who chose chocolate is greater than the number of players who chose vanilla , which in turn is greater than the number of players who chose strawberry . Let 's denote the number of players who chose chocolate , vanilla , and strawberry as (c) , (v) , and (s) respectively . We know that $(c > v > s)$ and $(c + v + s = 9)$ with (c) , (v) , and (s) being positive integers . We need to count the number of solutions to this equation under the given constraints .

We can start by finding the number of non-negative integer solutions to $(c + v + s = 9)$ without the constraints . This is given by the stars and bars method , which gives us $(\binom{9 + 2}{2} = \binom{11}{2} = 55)$ solutions . However , we need to subtract the cases where $(c \geq v > s)$.

To do this , we can reindex our variables . Let $(c' = c - 1)$, $(v' = v)$, and $(s' = s)$. Then we have $(c' + 1 + v' + s' = 9 \Rightarrow c' + v' + s' = 8)$ with $(c' \geq v' > s')$ and $(c' + v' + s' \geq 2)$ because $(c \geq v > s)$. Now we need to count the number of solutions to $(c' + v' + s' = 8)$ with $(c' \geq v' > s')$.

We can use a change of variables to simplify . Let $(c' = a + b)$, $(v' = a)$, and $(s' = c - a - b)$. Since $(c' > v' > s')$ and $(a \leq b)$, we need to count the number of pairs $((a , b))$ such that $(a \leq b \leq c' \leq \frac{8}{2} = 4)$ and $(c' \leq \frac{8}{3} \approx 2.67)$. So (c') can be 2 , 3 , or 4 .

- 1 . If $(c' = 2)$, then $(a = 1)$ and $(b = 1)$.
- 2 . If $(c' = 3)$, then $(a = 1)$ and $(b = 2)$.
- 3 . If $(c' = 4)$, then $(a = 1)$ and $(b = 3)$.

Each of these corresponds to a specific distribution of (c , v , s) :

- 1 . $(c = 3 , v = 3 , s = 2)$
- 2 . $(c = 2 , v = 3 , s = 3)$
- 3 . $(c = 1 , v = 3 , s = 5)$ (not valid because $(v \leq c)$)

So there are 2 valid distributions . Each of these can be permuted among the 9 players , giving us $(2! \times 2! \times 2! = 2^3 = 8)$ ways to assign the flavors . Thus , the total number of ways is $(8 \times 2 = 16)$.

Finally , we need the remainder when this number is divided by 1000 , which is (16) .

<answer>16</answer>